

BEHLOOL MOIZ

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SUMMARY

Electrical Engineering student with experience designing circuits and custom PCBs, integrating microcontrollers and sensors, running Multiphysics and circuit simulations. Seeking an engineering internship or co-op for Fall 2026, Spring or Summer 2027.

EDUCATION

University of Houston

Bachelor of Science in Electrical Engineering

May 2028

GPA: **3.63**

ENGINEERING & LEADERSHIP EXPERIENCE

Cullen College of Engineering, University of Houston | Houston, TX

June 2026 – Present

Circuit Analysis 2 Teaching Assistant (TA)

- Reduced grading time by 60% by developing a Python script that automated submission grading and custom feedback via an AI model API.
- Designed original homeworks/quizzes and conducted weekly office hours to troubleshoot complex AC network analysis, transfer functions, and transient responses.
- Increased section performance from a 60% historical average to a high 70% average as a direct result of weekly review sessions, effective grading, and feedback.

Bioelectronics Synergy Laboratory, University of Houston | Houston, TX

May 2026 – Aug 2026

Research Assistant (RA)

- Conducted literature reviews and COMSOL simulations on cell migration methods, analyzing how localized electric fields can be used to safely direct cells towards target zones.
- Designed custom PCBs and prototype device components using soft plastics like PDMS, GelMA hydrogels and conductive polymers (PEDOT: PSS)
- Facilitated weekly lab meetings with the research team to present project updates and integrate electrical hardware with biological systems.

Institute of Electrical and Electronics Engineers (IEEE)

Oct 2025 – May 2026

Project & Workshop Chair

- Led and organized hands-on electronics workshops (breadboarding, soldering, PCB design using EasyEDA) with 40+ attendees per session.
- Coordinated end-of-end workshop logistics, managing materials, schedules, and officer roles to support 15+ events per semester.
- Represented the university at an IEEE regional conference, traveling to compete against other universities in technical student circuit design competitions.

PROJECTS

AI Parking Space Detection System

Feb – March 2026

- Built a computer vision system using YOLOv8n trained on a 10-k-image dataset to detect parking occupancy from CCTV feeds.
- Achieved >99% accuracy on free vs. occupied classification with real-time inference for live lot monitoring.
- Designed a scalable backend to handle high request volumes, reducing search time by 10+ minutes in crowded areas.

Micromouse Autonomous Robot

Oct 2025 – Present

- Designed and made a custom PCB using KiCAD to control DC motors and integrate an IMU for navigation
- Programmed the embedded system in ESP32 using C++, implementing PWM for smooth motor control.
- Implemented PWM-driven movement and turn control with IMU-based heading correction for maze navigation.

Solar Greenhouse Recorder

Aug – Nov 2025

- Built a 3D-printed mini greenhouse powered by a 800mAh lithium battery and 8×8-inch solar panel.
- Programmed an ESP32-CAM to capture images every 60 seconds and store them on an SD card.
- Developed a Python script to compile the images into a time-lapse video, producing a continuous visual record of plant growth.

SKILLS & AWARDS

Dean's List: Fall 2024, Spring 2025, Fall 2025

Cullen Long Family Endowment and **S. Javaid Anvar** Scholarship recipient.

Certifications: Excel and Power BI (Hewlett Packard Enterprise DSI Badge),

Application Software: MATLAB, Python, C, EasyEDA, KiCAD, AutoCAD, SolidWorks, Power BI, Microsoft Office,

Languages: Multilingual (English, Hindi, Urdu & Gujarati), (A2 in Spanish)